

Introduction to Offensive Security

Ethical Hacking — Course Orientation

Welcome! Let's talk about what this course is, what it isn't, and the one rule that matters most.

What is "offensive security"?

- Testing systems by **attacking them — with permission** — to find weaknesses before criminals do.
- Also called **ethical hacking** or **penetration testing**.
- Core idea: **You can't protect what you don't understand.**

We learn how attackers think so we can become better **defenders**.

Why learn to attack?

- Every defender, developer, and IT pro is stronger if they understand offense.
- Huge, growing career field: pentester, SOC analyst, red/blue team, AppSec, GRC...
- It's genuinely fun: puzzles, problem-solving, "how does this *really* work?"

Hats: who's who

Hat	Who they are
 White hat	Hacks legally, with permission, to help
 Black hat	Criminal — no permission, causes harm
 Gray hat	No clear permission, no malice — still legally risky

Red team = attackers · **Blue team** = defenders



The one rule

Authorization is the **ONLY** line between a penetration tester and a criminal.

You may attack **only** systems you **own** or have **explicit written permission** to test, within a defined **scope**.

What that means here

- Everything we do happens in **safe, isolated lab environments** built to be attacked.
-   Never the school's systems.   Never classmates' devices.   Never a real website.
- Unauthorized access is **illegal** — Computer Fraud and Abuse Act (CFAA) + state law.
- **Minors are not exempt.** Real consequences: charges, discipline, lasting impact.

If you find a real vulnerability...

Do NOT exploit it, share it, or post it.

DO report it privately to the owner (or your teacher). That's called **responsible disclosure** — and companies even pay for it through **bug bounty** programs.

Report, don't exploit. That habit defines a professional.

How the course works

- **6 modules → 18 units**, roughly one unit per week.
- Each week: learn the concept → do a **hands-on lab** → document it in your **lab journal**.
- Tools the pros use (Kali, nmap, Burp, Metasploit) against **safe practice targets**.
- Platforms: **TryHackMe / HTB Academy**, **local VirtualBox VMs**, **picoCTF / OverTheWire**.

The roadmap

- 1. Foundations & ethics** → build a safe lab
- 2. Technical foundations** → networking, Linux, Bash, Python
- 3. Reconnaissance** → find information & scan
- 4. Exploitation** → web attacks, SQLi, exploits, passwords
- 5. Post-exploitation** → privilege escalation, Metasploit
- 6. Putting it together** → reporting + capstone CTF + careers

A note on using AI

Use of Claude and ChatGPT is **encouraged** — but make sure **you** understand the content instead of blindly telling it what to do.

AI is powerful in cybersecurity and isn't going away. Use it as a **tutor** (explain a concept, debug a script, check your thinking) — never as an **autopilot** that does the learning for you.

Before we touch a single tool

- Read and **sign the Acceptable Use & Ethics Agreement** (you *and* a parent/guardian).
- We'll set up your lab next unit and verify it's **isolated** from the real internet.
- Bring your curiosity — and your judgment.

Let's get started

Unit 1: What Is Offensive Security? Ethics, Law & the Hacker Mindset

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